

Geometric Quantities for the Schwarzschild Metric (with Weyl Tensor)

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1 Citation Information

... (same as before) ...

2 Geometric quantities for the Schwarzschild metric

The Schwarzschild metric is given by

$$ds^2 = - \left(1 - \frac{2GM}{c^2 r} \right) c^2 dt^2 + \left(1 - \frac{2GM}{c^2 r} \right)^{-1} dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2. \quad (1)$$

2.1 Metric tensor $g_{\mu\nu}$

$$\begin{bmatrix} -c^2 \left(-\frac{R_s}{r} + 1 \right) & 0 & 0 & 0 \\ 0 & \frac{1}{-\frac{R_s}{r} + 1} & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ 0 & 0 & 0 & r^2 \sin^2(\theta) \end{bmatrix} \quad (2)$$

2.2 Christoffel symbols $\Gamma_{\nu\rho}^{\mu}$ (non-zero)

$$\Gamma_{tr}^t = \frac{R_s}{2r(-R_s + r)} \quad (3)$$

$$\Gamma_{rt}^t = \frac{R_s}{2r(-R_s + r)} \quad (4)$$

$$\Gamma_{tt}^r = \frac{R_s c^2 (-R_s + r)}{2r^3} \quad (5)$$

$$\Gamma_{rr}^r = \frac{R_s}{2r(R_s - r)} \quad (6)$$

$$\Gamma_{\theta\theta}^r = R_s - r \quad (7)$$

$$\Gamma_{\phi\phi}^r = (R_s - r) \sin^2(\theta) \quad (8)$$

$$\Gamma_{r\theta}^{\theta} = \frac{1}{r} \quad (9)$$

$$\Gamma_{\theta r}^{\theta} = \frac{1}{r} \quad (10)$$

$$\Gamma_{\phi\phi}^{\theta} = -\frac{\sin(2\theta)}{2} \quad (11)$$

$$\Gamma_{r\phi}^{\phi} = \frac{1}{r} \quad (12)$$

$$\Gamma_{\theta\phi}^{\phi} = \frac{1}{\tan(\theta)} \quad (13)$$

$$\Gamma_{\phi r}^{\phi} = \frac{1}{r} \quad (14)$$

$$\Gamma_{\phi\theta}^{\phi} = \frac{1}{\tan(\theta)} \quad (15)$$

$$(16)$$

2.3 Riemann curvature tensor $R^\mu_{\nu\rho\sigma}$ (non-zero)

$$R^t_{rtr} = \frac{R_s}{r^2 (-R_s + r)} \quad (17)$$

$$R^t_{rrt} = \frac{R_s}{r^2 (R_s - r)} \quad (18)$$

$$R^t_{\theta t \theta} = -\frac{R_s}{2r} \quad (19)$$

$$R^t_{\theta \theta t} = \frac{R_s}{2r} \quad (20)$$

$$R^t_{\phi t \phi} = -\frac{R_s \sin^2(\theta)}{2r} \quad (21)$$

$$R^t_{\phi \phi t} = \frac{R_s \sin^2(\theta)}{2r} \quad (22)$$

$$R^r_{ttr} = \frac{R_s c^2 (-R_s + r)}{r^4} \quad (23)$$

$$R^r_{trt} = \frac{R_s c^2 (R_s - r)}{r^4} \quad (24)$$

$$R^r_{\theta r \theta} = -\frac{R_s}{2r} \quad (25)$$

$$R^r_{\theta \theta r} = \frac{R_s}{2r} \quad (26)$$

$$R^r_{\phi r \phi} = -\frac{R_s \sin^2(\theta)}{2r} \quad (27)$$

$$R^r_{\phi \phi r} = \frac{R_s \sin^2(\theta)}{2r} \quad (28)$$

$$R^\theta_{tt\theta} = \frac{R_s c^2 (R_s - r)}{2r^4} \quad (29)$$

$$R^\theta_{t\theta t} = \frac{R_s c^2 (-R_s + r)}{2r^4} \quad (30)$$

$$R^\theta_{rr\theta} = \frac{R_s}{2r^2 (-R_s + r)} \quad (31)$$

$$R^\theta_{r\theta r} = \frac{R_s}{2r^2 (R_s - r)} \quad (32)$$

$$R^\theta_{\phi \theta \phi} = \frac{R_s \sin^2(\theta)}{r} \quad (33)$$

$$R^\theta_{\phi \phi \theta} = -\frac{R_s \sin^2(\theta)}{r} \quad (34)$$

$$R^\phi_{tt\phi} = \frac{R_s c^2 (R_s - r)}{2r^4} \quad (35)$$

$$R^\phi_{t\phi t} = \frac{R_s c^2 (-R_s + r)}{2r^4} \quad (36)$$

$$R^\phi_{rr\phi} = \frac{R_s}{2r^2 (-R_s + r)} \quad (37)$$

$$R^\phi_{r\phi r} = \frac{3 R_s}{2r^2 (R_s - r)} \quad (38)$$

$$R^\phi_{\theta \theta \phi} = -\frac{R_s}{r} \quad (39)$$

$$R^\phi_{\theta \phi \theta} = \frac{R_s}{r} \quad (40)$$

$$(41)$$

2.4 Weyl curvature tensor $C^\mu_{\nu\rho\sigma}$ (non-zero)

$$C^t_{rtr} = -\frac{R_s c^2}{r^3} \quad (42)$$

$$C^t_{rrt} = \frac{R_s c^2}{r^3} \quad (43)$$

$$C^t_{\theta t \theta} = \frac{R_s c^2 (-R_s + r)}{2r^2} \quad (44)$$

$$C^t_{\theta \theta t} = \frac{R_s c^2 (R_s - r)}{2r^2} \quad (45)$$

$$C^t_{\phi t \phi} = \frac{R_s c^2 (-R_s + r) \sin^2(\theta)}{2r^2} \quad (46)$$

$$C^t_{\phi \phi t} = \frac{R_s c^2 (R_s - r) \sin^2(\theta)}{2r^2} \quad (47)$$

$$C^r_{ttr} = \frac{R_s c^2}{r^3} \quad (48)$$

$$C^r_{trt} = -\frac{R_s c^2}{r^3} \quad (49)$$

$$C^r_{\theta r \theta} = \frac{R_s}{2(R_s - r)} \quad (50)$$

$$C^r_{\theta \theta r} = \frac{R_s}{2(-R_s + r)} \quad (51)$$

$$C^r_{\phi r \phi} = \frac{R_s \sin^2(\theta)}{2(R_s - r)} \quad (52)$$

$$C^r_{\phi \phi r} = \frac{R_s \sin^2(\theta)}{2(-R_s + r)} \quad (53)$$

$$C^\theta_{tt\theta} = \frac{R_s c^2 (R_s - r)}{2r^2} \quad (54)$$

$$C^\theta_{t\theta t} = \frac{R_s c^2 (-R_s + r)}{2r^2} \quad (55)$$

$$C^\theta_{rr\theta} = \frac{R_s}{2(-R_s + r)} \quad (56)$$

$$C^\theta_{r\theta r} = \frac{R_s}{2(R_s - r)} \quad (57)$$

$$C^\theta_{\phi\theta\phi} = R_s r \sin^2(\theta) \quad (58)$$

$$C^\theta_{\phi\phi\theta} = -R_s r \sin^2(\theta) \quad (59)$$

$$C^\phi_{tt\phi} = \frac{R_s c^2 (R_s - r) \sin^2(\theta)}{2r^2} \quad (60)$$

$$C^\phi_{t\phi t} = \frac{R_s c^2 (-R_s + r) \sin^2(\theta)}{2r^2} \quad (61)$$

$$C^\phi_{rr\phi} = \frac{R_s \sin^2(\theta)}{2(-R_s + r)} \quad (62)$$

$$C^\phi_{r\phi r} = \frac{R_s \sin^2(\theta)}{2(R_s - r)} \quad (63)$$

$$C^\phi_{\theta\theta\phi} = -R_s r \sin^2(\theta) \quad (64)$$

$$C^\phi_{\theta\phi\theta} = R_s r \sin^2(\theta) \quad (65)$$

$$(66)$$

2.5 Ricci tensor $R_{\mu\nu}$ (non-zero)

$$\text{All components zero.} \quad (67)$$

2.6 Ricci scalar R

$$0 \quad (68)$$

2.7 Einstein tensor $G_{\mu\nu}$ (non-zero)

$$\text{All components zero.} \quad (69)$$

2.8 Geodesic equations

$$\ddot{t} = -\frac{R_s \dot{t}}{r(-R_s + r)} \quad (70)$$

$$\ddot{r} = -\frac{R_s \dot{r}^2}{2r(R_s - r)} - \frac{R_s \dot{t}^2 c^2 (-R_s + r)}{2r^3} - \dot{\phi}^2 (R_s - r) \sin^2(\theta) - \dot{\theta}^2 (R_s - r) \quad (71)$$

$$\ddot{\theta} = \frac{\dot{\phi}^2 \sin(2\theta)}{2} - \frac{2\dot{\theta}\dot{r}}{r} \quad (72)$$

$$\ddot{\phi} = -\frac{2\dot{\phi}\dot{\theta}}{\tan(\theta)} - \frac{2\dot{\phi}\dot{r}}{r} \quad (73)$$

$$(74)$$

where λ is an affine parameter and $\dot{x}^\mu = dx^\mu/d\lambda$.

2.9 Kretschmann scalar $R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$

Full expression

$$\frac{R_s^2 c^4 (-R_s + r)^2 \left(-\frac{R_s}{r} + 1\right)^2}{r^4 (R_s - r)^2 (-R_s c^2 + c^2 r)^2} + \frac{2R_s^2 c^4 \left(-\frac{R_s}{r} + 1\right)^2}{r^4 (-R_s c^2 + c^2 r)^2} + \frac{R_s^2 c^4 (-R_s + r)^2}{2r^6 (-R_s c^2 + c^2 r)^2} + \frac{R_s^2 c^4 (R_s - r)^2}{2r^6 (-R_s c^2 + c^2 r)^2} + \frac{R_s^2 c^4 (R_s - r)^2}{r^8 (-R_s c^2 + c^2 r)^2} \quad (75)$$

Simplified expression

$$\frac{12R_s^2}{r^6} \quad (76)$$

2.10 Ricci squared $R_{\mu\nu}R^{\mu\nu}$

$$0 \tag{77}$$

2.11 Gauss-Bonnet scalar $\mathcal{R}_{GB} = R^2 - 4R_{\mu\nu}R^{\mu\nu} + R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$

$$\frac{12R_s^2}{r^6} \tag{78}$$

2.12 Weyl squared scalar $C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma}$

$$\frac{3R_s^2 \left(c^4 (R_s - r)^4 + r^6 (R_s - r)^2 (\sin^4(\theta) + 1) + r^4 \right)}{r^8 (R_s - r)^2} \tag{79}$$